Documentation

Supply Chain Visibility System with Optimization Analytics

Tools Used:	Microsoft Power BI, Google Colab
Dataset:	AIS Data Complete Weather
Milestones:	1 – 4
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Milestone 1 – Data Modelling & KPI Foundation

The project titled **"Supply Chain Visibility System with Optimization Analytics"** focuses on using data analytics to improve the visibility of supply-chain transportation activities.

Supply chains involve the movement of goods from one location to another. Monitoring transportation activities such as vessel location, speed, destination, estimated arrival time and weather conditions can help organizations understand transportation operations and identify areas for improvement [3].

For this project, an **AIS (Automatic Identification System)** dataset was used. AIS is a system used by ships to transmit information such as vessel identification, location, speed, course and other operational information [1, 2].

The raw dataset was first cleaned in Google Colab, and Microsoft Power BI was then used to transform the cleaned dataset into an interactive analytical dashboard.

Dataset Description

The dataset used for this milestone is **AIS Data Complete Weather** [4].

The dataset contains approximately **1.1 million** AIS observations (1,098,937 rows in the cleaned `Fact_Vessel` table) and information about approximately **13,000** unique vessels.

Each row represents an observation of a vessel at a particular point in time. Some important fields in the dataset are:

Field	Description
`MMSI`	Unique identification number of a vessel
`VesselName`	Name of the vessel
`BaseDateTime`	Date and time of the observation
`LAT`	Latitude of the vessel
`LON`	Longitude of the vessel
`SOG`	Vessel speed
`SOG_kmh`	Vessel speed in kilometres per hour
`VesselType`	Vessel type / category code
`Status`	Vessel operational / navigation status
`Cargo`	AIS cargo classification / code
`dist_km`	Distance-related information
`ETA_hours`	Estimated time of arrival
`dest_cluster`	Destination grouping
`weather_wind_kph`	Wind speed
`weather_visibility_km`	Visibility
`weather_temp_c`	Temperature

Table 1: Important fields in the AIS dataset

The dataset combines vessel movement information with weather information, making it useful

for transportation and supply-chain visibility analysis. The field definitions are consistent with the general AIS data context described in the AIS data resources [1, 2].

Data Preparation

The raw dataset was first cleaned in Google Colab. The cleaned dataset was then imported into Microsoft Power BI and prepared using Power Query, a data preparation and transformation tool [9, 7].

Data Type Checking

The data types of important columns were checked and adjusted where required, for example:

- `MMSI` → identification field
- `BaseDateTime` → date/time
- `LAT` and `LON` → numerical
- `SOG` and `SOG_kmh` → numerical
- `ETA_hours` → numerical
- Weather fields → numerical

Data Transformation

The dataset was transformed to make it easier to analyze in Power BI. Two derived fields were created:

Operational_Status – categorizes vessel observations into operational states such as:

- Moving
- Stopped

Speed_Category – vessel speed was grouped into categories:

- Stopped
- Slow
- Moderate
- Fast

These categories make the dashboard easier to understand than displaying only raw speed values.

The cleaned dataset as loaded into Power BI (`Fact_Vessel` table, 1,098,937 rows) is shown below.

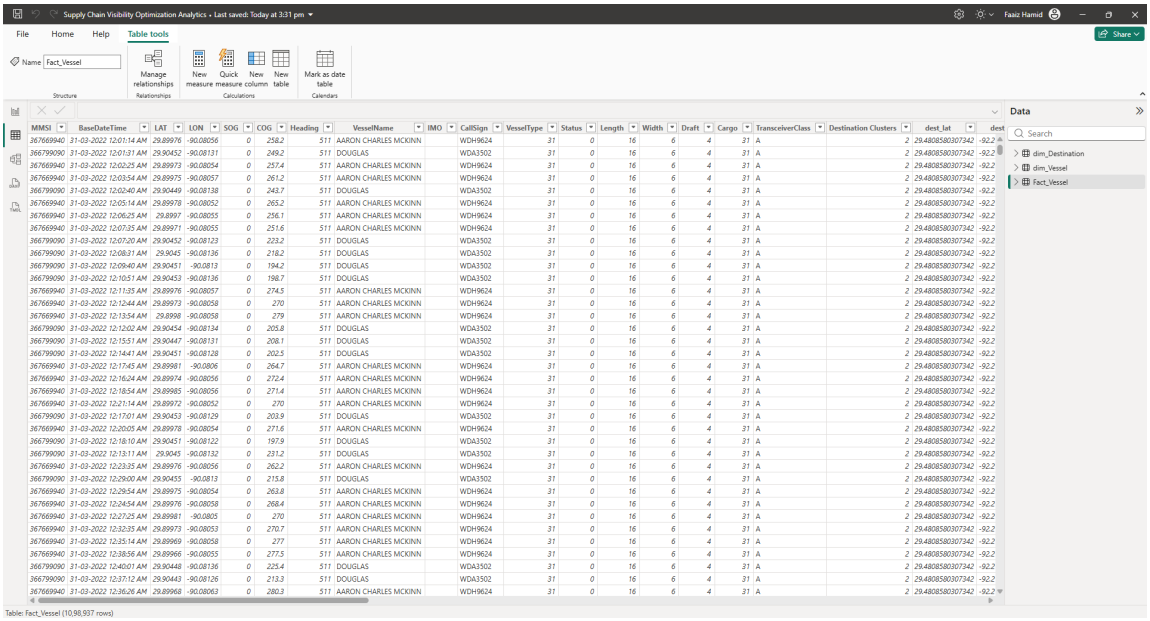


Figure 1: Cleaned dataset loaded into Power BI (Fact_Vessel table)

Data Model

A simple dimensional data model was created in Power BI. Dimensional modelling and star-schema structures are commonly used to organize analytical data for efficient filtering and aggregation [5, 8]. The main tables used were:

- **Fact_Vessel**
- **dim_Vessel**
- **dim_Destination**

Fact_Vessel. Acts as the central table containing the main AIS observations. It contains information such as:

- Vessel
- Time
- Location
- Speed
- Destination
- ETA
- Weather
- Operational status

dim_Vessel. Provides descriptive information about vessels (such as MMSI, VesselName, VesselType, CallSign, IMO, Length, Draft, Cargo and TransceiverClass) and allows vessel-related filtering and analysis.

dim_Destination. Provides destination-related information (`dest_cluster` with its `dest_lat` and `dest_lon` coordinates) and allows the data to be grouped according to destination clusters.

Relationships. The model follows a simple star-schema-style structure. One vessel can have multiple AIS observations. Therefore, the relationship between the vessel dimension and the fact table is based on a **one-to-many** relationship. This structure makes filtering and aggregation easier within Power BI, in line with Microsoft's star-schema guidance [8]. Similarly, each destination cluster can be linked to many AIS observations, so `dim_Destination` is also related to `Fact_Vessel` through a one-to-many relationship.

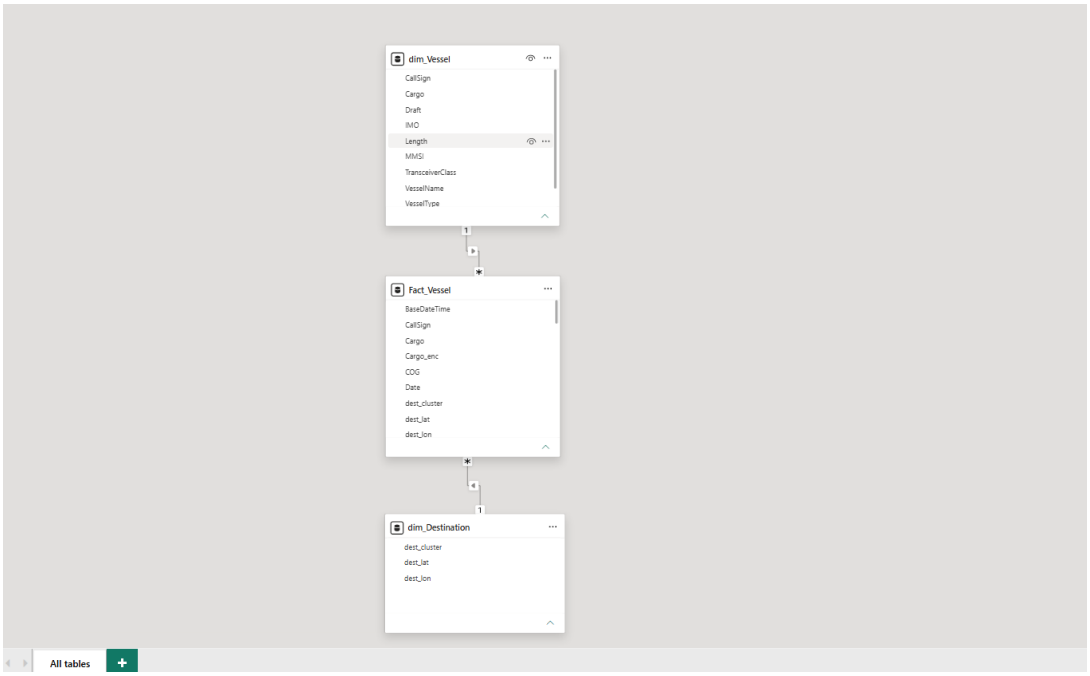


Figure 2: Star-schema data model in Power BI

KPI Development

KPIs, or Key Performance Indicators, are important summary values used to understand the overall condition of the data. Several DAX measures were created in Power BI using Data Analysis Expressions (DAX), Microsoft's formula language for analytical calculations [6, 10].

Total AIS Records
Measures the total number of observations in the dataset.

Unique Vessels
Calculates the number of different vessels.

Average Speed
Calculates the average vessel speed.

Total Distance
Calculates the total distance represented by the available records.

Average ETA
Calculates the average estimated arrival time.

Stopped Records
Calculates the number of observations where the vessel was classified as stopped.

Stopped AIS Records %
Calculates the percentage of AIS observations classified as stopped.

Dashboard – Page 1: Supply Chain Visibility

The first dashboard page, "Supply Chain Visibility – Vessel Movement and Operational Performance", provides an overall view of vessel and transportation activity. The page contains three slicers, six KPI cards and four charts.

KPI Cards

The dashboard contains the following major KPIs:

KPI	Result
Total AIS Records	~1.1 Million
Unique Vessels	~13,000
Average Speed	4.36 km/h
Moving AIS Records	224K
Average ETA	59.89 hours
Stopped AIS Records %	79.64%

Table 2: Page 1 KPI cards

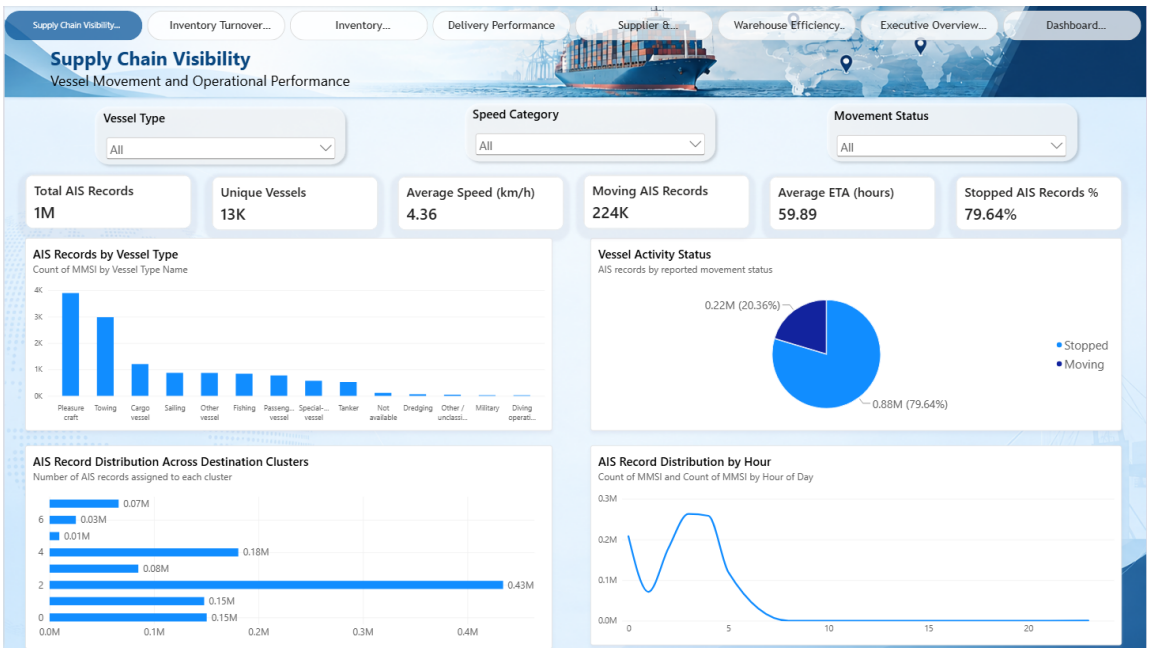


Figure 3: Page 1 – Supply Chain Visibility dashboard

KPIs provide a quick summary of the transportation activity represented in the dataset.

Dashboard Visualizations

AIS Records by Vessel Type. This chart shows the count of MMSI by vessel type name. Pleasure craft, Towing and Cargo vessel are the most represented categories. It helps identify which vessel categories are most represented in the dataset.

Vessel Activity Status. A pie chart shows the proportion of AIS records by reported movement status (Stopped or Moving). The dashboard shows approximately:

- 79.64% Stopped (0.88M records)
- 20.36% Moving (0.22M records)

This indicates that a large proportion of the recorded observations were classified as stopped.

AIS Record Distribution Across Destination Clusters. This chart shows the number of AIS records assigned to each destination cluster. Cluster 2 has the highest volume (approximately 0.43M), followed by cluster 4 (approximately 0.18M) and clusters 0 and 1 (approximately 0.15M each), while cluster 5 has the fewest (approximately 0.01M).

AIS Record Distribution by Hour. A line chart shows the number of AIS records by hour of day. Record volume peaks at approximately 0.26M around hours 3–4, dips to approximately 0.07M around hour 1, and falls to near zero from about hour 8 onwards.

Interactive Slicers

Interactive slicers were added to make the dashboard easier to explore. The main slicers include:

- Vessel Type
- Speed Category
- Movement Status

When a user selects a particular value, the relevant charts and KPI cards automatically update. For example, selecting one vessel type allows the user to analyze only the activity associated with that vessel category. This makes the dashboard interactive rather than static.

Dashboard – Page 2: Inventory Turnover Analysis

The second dashboard page focuses on cargo-flow and inventory-turnover-related operational analysis.

The available AIS dataset does not contain conventional inventory information such as:

- Inventory quantity
- Cost of goods sold (COGS)
- Average inventory value

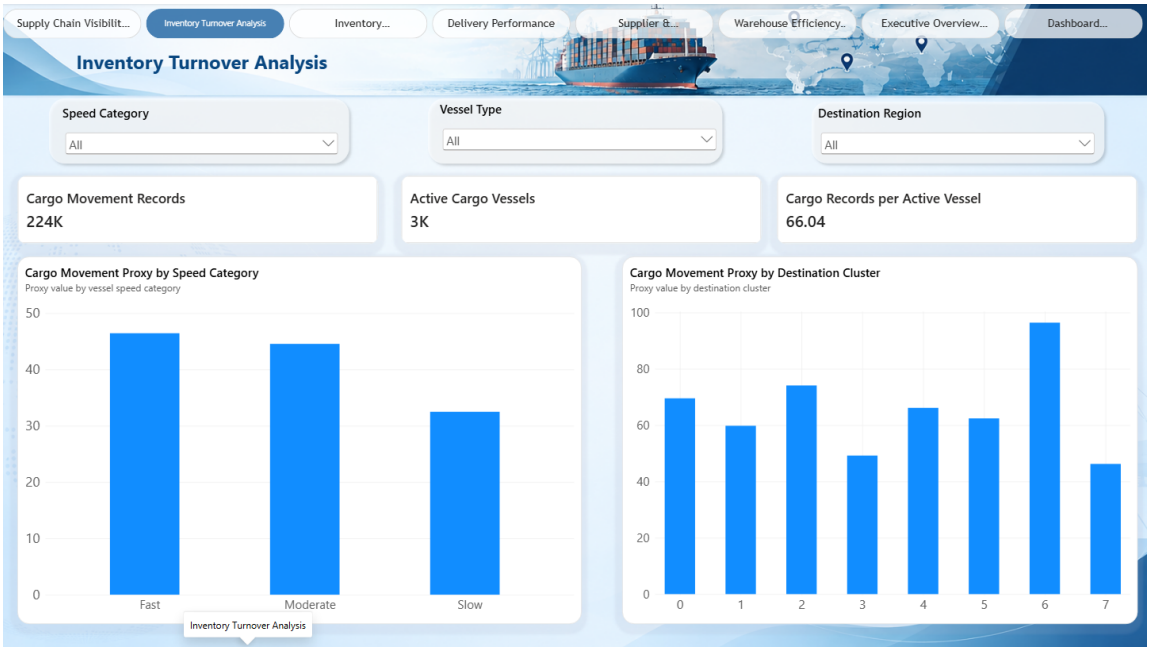


Figure 4: Page 2 – Inventory Turnover Analysis dashboard

Therefore, a standard accounting inventory turnover ratio could not be calculated directly from this dataset. Instead, a **Cargo Movement Proxy** was developed.

Cargo Movement Proxy

Three measures were created for the operational analysis:

- **Cargo Movement Records**
- **Active Cargo Vessels**
- **Cargo Records per Active Vessel**

The resulting proxy value in the dashboard is approximately **66.04**. This value should not be interpreted as an accounting inventory turnover ratio. Instead, it represents an operational cargo-flow indicator (the average number of cargo movement records per active vessel), calculated by dividing Cargo Movement Records by Active Cargo Vessels. This approach was used because the supplied AIS dataset does not contain the financial/inventory fields required for a conventional inventory turnover calculation; therefore, the measure is explicitly treated as a project-defined operational proxy rather than a standard accounting ratio.

Page 2 Visualizations

The second page contains:

Cargo Movement Records Card. Shows the number of AIS observations associated with moving vessels.

Active Cargo Vessels Card. Shows the number of unique vessels classified as moving (approximately 3K).

Cargo Records per Active Vessel Card. Displays the calculated operational proxy value (66.04).

Cargo Movement Proxy by Speed Category. Shows how the proxy varies across vessel speed categories: approximately 46 for Fast, 44 for Moderate and 33 for Slow.

Cargo Movement Proxy by Destination Cluster. Shows the proxy across the eight destination clusters, from approximately 46 (cluster 7) to approximately 96 (cluster 6).

The page can be filtered using three slicers: Speed Category, Vessel Type and Destination Region. These visualizations provide another perspective on vessel activity and cargo movement.

Key Dashboard Results

The major results obtained from the dashboard are:

- Approximately 1.1 million AIS observations were analyzed.
- Approximately 13,000 unique vessels were identified.
- The average vessel speed was approximately 4.36 km/h.
- The total recorded distance was approximately 277.87 million km.
- The average ETA was approximately 59.89 hours.
- Approximately 79.64% of observations were classified as stopped.
- Approximately 20.36% were classified as moving.
- The operational cargo-movement proxy (cargo records per active vessel) was approximately 66.04, based on approximately 224K cargo movement records and approximately 3K active cargo vessels.

These results provide a high-level understanding of the transportation activity contained in the dataset.

Business Insights

1. **Vessel Activity** – The dashboard provides an overview of the number and types of vessels represented in the dataset.
2. **Movement Status** – The stopped/moving analysis provides visibility into vessel operational status.
3. **Transportation Speed** – Speed categories make it easier to identify different levels of vessel movement.
4. **Destination Activity** – Destination clusters help identify areas with relatively higher AIS activity.
5. **ETA Monitoring** – Average ETA provides an overview of expected arrival duration.
6. **Weather Context** – Combining weather information with vessel movement provides additional context for transportation analysis.

7. **Cargo Flow** – The operational cargo-movement proxy (cargo records per active vessel) provides an additional indicator of cargo-flow activity when conventional inventory data is unavailable.

Tools and Technologies Used

Microsoft Power BI. Used for:

- Data import
- Data transformation
- Data modelling
- DAX calculations
- Dashboard development
- Interactive filtering
- Data visualization

Google Colab. Used for:

- Cleaning the raw AIS dataset before it was imported into Power BI

Power Query. Used for:

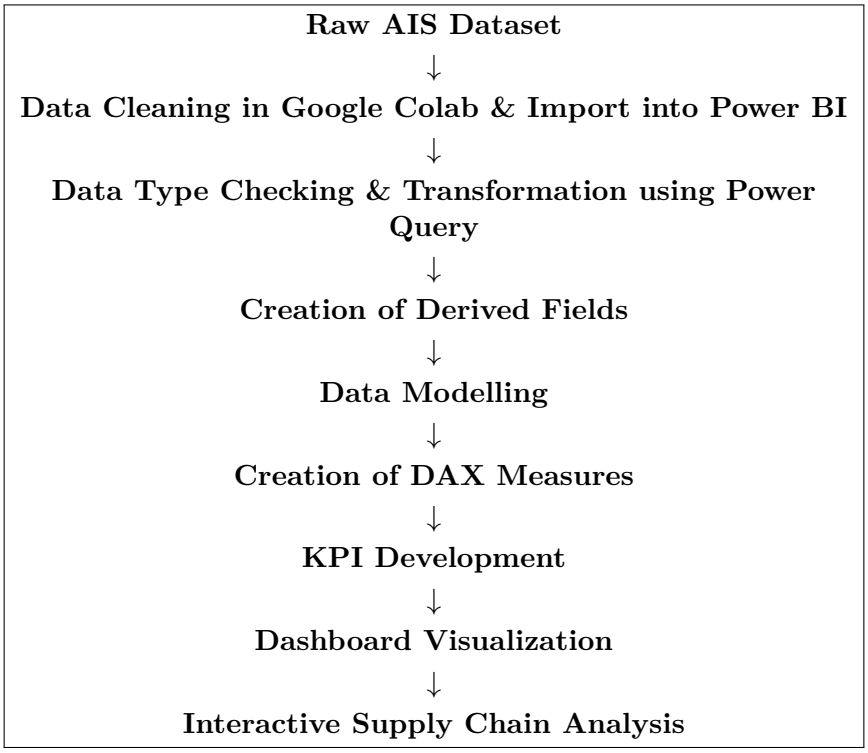
- Data preparation
- Data type checking
- Data transformation
- Creating derived fields

DAX. DAX (Data Analysis Expressions) was used to create calculated measures and KPIs, following the calculation capabilities documented for DAX [6, 10]. Examples include:

- Total AIS Records
- Unique Vessels
- Average Speed
- Total Distance
- Average ETA
- Stopped %
- Cargo Records per Active Vessel

Milestone 1 Workflow

The overall workflow followed during Milestone 1 was:


Raw AIS Dataset
↓
Data Cleaning in Google Colab & Import into Power BI
↓
Data Type Checking & Transformation using Power Query
↓
Creation of Derived Fields
↓
Data Modelling
↓
Creation of DAX Measures
↓
KPI Development
↓
Dashboard Visualization
↓
Interactive Supply Chain Analysis

Milestone 2 – Inventory & Delivery Analytics

Dashboard – Page 3: Inventory Performance

This page presents *Cargo Flow Analysis with Regional and Product-Level Insights*, focusing on cargo movement volume, vessel activity and cargo records per active vessel across speed categories.

KPI Cards

KPI	Result
Cargo Movement Records	224K
Unique Vessels	13K
Average ETA (hours)	59.89
Cargo Records per Active Vessel	66.04

Table 3: Page 3 KPI cards

Cargo Records per Active Vessel by Speed Category

A column chart shows the average number of cargo movement records per active vessel, grouped by `Speed_Category`. Fast-moving vessels show the highest value (approximately 46), followed by Moderate (approximately 44) and Slow (approximately 33), indicating that vessels moving at higher speeds are associated with higher cargo-flow turnover.

Cargo Movement Records by Speed Category

A donut chart shows how the 224K cargo movement records are distributed across speed categories: Moderate accounts for 100K records (44.58%), Slow for 84K (37.33%) and Fast for 40K (18.09%).

Interactive Slicers

The page includes the following slicers, allowing the report to be filtered dynamically:

- Vessel Type (`VesselType`)
- Speed Category (`Speed_Category`)
- Destination Region (`dest_cluster`)

Dashboard – Page 4: Delivery Performance

This page presents *Regional & Product-Level Analysis with Trend and Variance*, focusing on on-time delivery performance, estimated arrival time (ETA) trends by hour, and ETA breakdowns by destination cluster, movement status and vessel type.

KPI Cards

KPI	Result
On-Time Delivery %	25.56%
Delayed Deliveries	818K
Average ETA (hours)	59.89
Count of Destination Clusters	8
ETA Variance	0.00

Table 4: Page 4 KPI cards

Estimated Arrival Time by Hour

A line chart plots the average estimated arrival time (hours) by hour of day. ETA remains stable at around 60 hours through the early hours, rises to a peak of approximately 72 hours around hour 8, then declines to approximately 60 hours around hour 15 and approximately 36 hours around hour 21, before rising slightly to approximately 39 hours at hour 23. This highlights time-of-day variance in estimated arrival time.

Average ETA by Destination Cluster

A column chart shows average ETA (hours) by destination cluster across the eight destination regions (0–7). ETA varies from approximately 28 hours (cluster 5, the lowest) to approximately 64 hours (clusters 4 and 7, the highest), indicating regional differences in delivery performance.

Average ETA by Movement Status

A column chart compares average ETA for stopped and moving vessel records. Stopped records show a considerably higher average ETA (approximately 70 hours) than moving records (approximately 23 hours).

Average ETA by Vessel Type

A column chart shows average ETA by vessel type, ranging from approximately 66 hours for Pleasure craft (the highest) to approximately 44 hours for Tanker (the lowest), providing a vessel-type-level view of delivery performance.

Interactive Slicers

The page includes the following slicers, allowing the report to be filtered dynamically:

- Speed Category (`Speed_Category`)
- Date
- Vessel Type (`VesselType`)

Key Dashboard Results

- Cargo movement records (moving observations) totalled approximately 224K.
- Approximately 13,000 unique vessels were recorded, of which approximately 3,000 were active cargo (moving) vessels.
- Average ETA was approximately 59.89 hours.
- Cargo records per active vessel (the operational turnover proxy) was approximately 66.04, varying by speed category (approximately 33–46) and by destination cluster (approximately 46–96).
- On-time delivery stood at approximately 25.56%, with approximately 818K delayed deliveries.
- Average ETA by hour peaked at approximately 72 hours around hour 8 and dipped to approximately 36 hours around hour 21.
- Average ETA by destination cluster ranged from approximately 28 hours (cluster 5) to approximately 64 hours (cluster 4) across the 8 destination clusters.
- Average ETA was approximately 70 hours for stopped records versus approximately 23 hours for moving records.
- ETA variance was recorded at 0.00 for the current filter context.

Business Insights

1. **Cargo Flow by Speed** – Higher vessel speed categories are associated with higher cargo records per active vessel, suggesting faster-moving vessels contribute more to cargo throughput.
2. **Regional Variation** – Both the cargo-movement proxy and ETA vary noticeably across destination clusters, pointing to specific regions that may need closer operational attention.
3. **Delivery Timing Patterns** – The hourly ETA trend reveals a peak in estimated arrival time around hour 8, useful for identifying potential congestion windows.
4. **On-Time Performance** – An on-time delivery rate of 25.56%, alongside 818K delayed deliveries, highlights a significant opportunity area for delivery performance improvement in later milestones.
5. **Movement Status and Vessel Type** – Average ETA is considerably higher for stopped records (approximately 70 hours) than for moving records (approximately 23 hours) and also varies across vessel types, suggesting that movement status and vessel type are relevant factors when reviewing arrival-time performance.

Dashboard Output

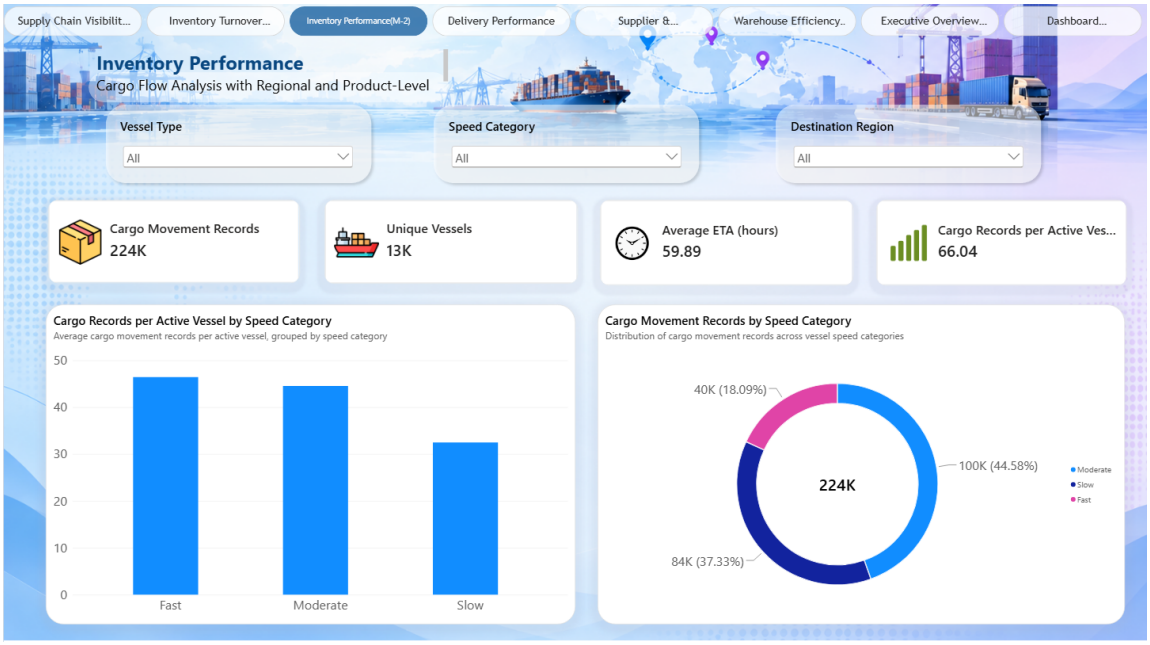


Figure 5: Page 3 – Inventory Performance dashboard

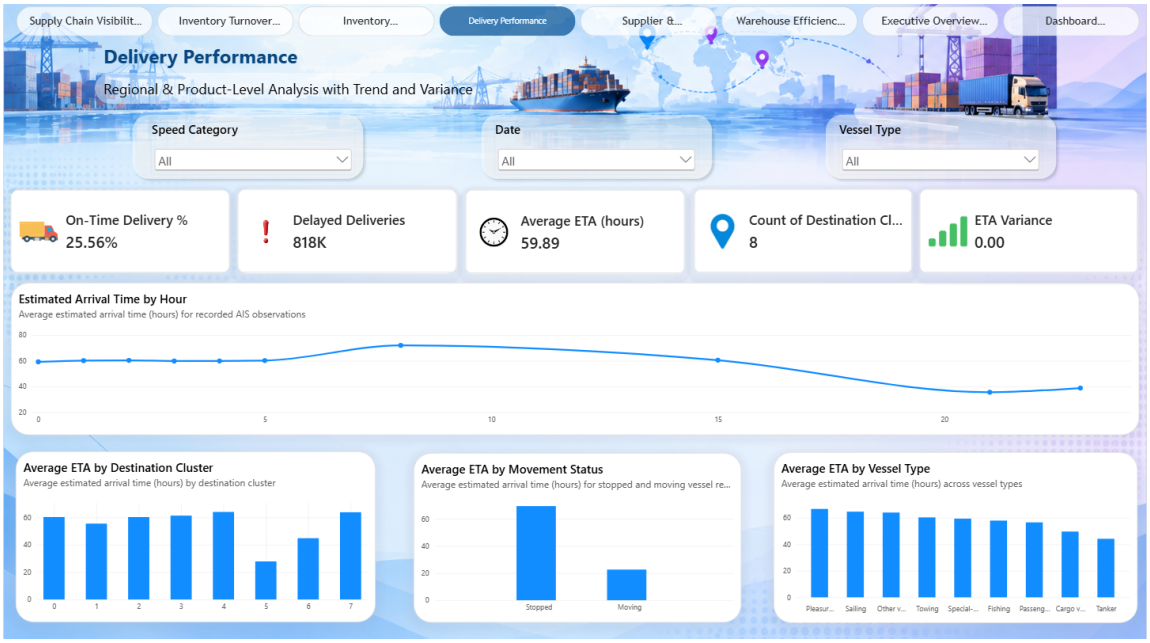


Figure 6: Page 4 – Delivery Performance dashboard

Milestone 3 – Supplier & Transportation Analytics

This milestone builds on the work completed so far in the project. The earlier milestones focused primarily on overall supply-chain visibility, inventory-related analysis, and delivery performance analysis.

Having established that foundation, this milestone extends the analysis to a new area of the supply chain – supplier and transportation activity – using the same underlying AIS (Automatic Identification System) dataset used throughout the project.

The main objective of Milestone 3 is to analyze:

- Supplier-level activity
- Transportation performance
- Carrier performance
- Route-wise transportation distribution
- Overall fulfillment and on-time performance

This provides a consolidated analytical view of the supplier and transportation side of the supply chain, complementing the inventory and delivery analysis produced in the previous milestones.

Dashboard Structure

The Milestone 3 dashboard, "Supplier & Transportation Analytics – Supplier Scorecards & Transportation Analysis" (Page 5), is organized into three sections:

- **Slicers (top)** – four interactive filters (Supplier, Region, Carrier, Route) that filter the entire dashboard.
- **KPI Cards (middle)** – four key performance indicators summarizing overall transportation and fulfillment performance.
- **Visualizations (bottom)** – four charts covering supplier activity, route-wise transportation cost, AIS records by vessel type, and supplier quality.

The intended usage flow is for a user to first review the high-level KPIs, then drill down using the charts and slicers for more detailed analysis.

KPI Cards

KPI	Result
Total Transportation Cost Proxy	277.87M
Average Shipment Cost Proxy	252.86
Fulfillment Rate	20.36%
On-Time Rate	25.56%

Table 5: Page 5 KPI cards

Fulfillment Rate note: since the dataset lacks a conventional order-fulfillment field, this figure should be read as an operational fulfillment / movement proxy rather than a traditional fulfillment percentage.

On-Time Rate note: this is an ETA-based indicator; records with an ETA above a defined threshold are treated as delayed.

Visualizations

Supplier-wise Shipment Count. Since a conventional supplier-name field is not available, Vessel Name is used as a supplier proxy. The chart shows the top 10 vessel names by record count, showing that some vessels have a considerably larger operational presence than others. In a real supplier dataset, this same approach could help identify high-volume suppliers for further review.

Route-wise Transportation Cost. A donut chart shows how the transportation-distance proxy is distributed across destination clusters, with the center displaying the overall total. The largest segment accounts for approximately 108.39M (≈39%) of total transportation distance, with the remaining clusters contributing smaller shares. This helps visualize how transportation activity is distributed geographically and highlights destinations with disproportionately high activity.

AIS Records by Vessel Type (Carrier Proxy). Since the dataset lacks a dedicated carrier field, Vessel Type is used as a carrier proxy. The chart shows the count of recorded AIS observations by vessel type code; activity is unevenly distributed, with vessel type 31 (approximately 0.4M records) and type 37 (approximately 0.2M) contributing significantly more than the others. If real carrier and cost data were available, a similar visualization could be used to compare carriers on cost and performance.

Supplier-wise Quality Score. Because no real supplier quality score exists in the dataset, an operational quality-score proxy based on ETA performance was created: lower average ETA results in a higher score, and the score is normalized to a 0–100 scale. This is explicitly a simplified operational proxy, not a genuine supplier-quality rating – a real system would incorporate factors such as delivery accuracy, product quality, rejection rate, lead time, and service-level compliance. In the dashboard view, the vessels shown all have a score of 100.

Interactive Slicers

Slicer	Based On	Purpose
Supplier	Vessel Name (supplier proxy)	Filter by supplier
Region	Destination Cluster	Filter by destination region
Carrier	Vessel Type (carrier proxy)	Filter by carrier / vessel type
Route	Destination Cluster	Filter by route (no separate route field available)

Table 6: Page 5 slicers

These slicers make the dashboard interactive – for example, selecting a specific vessel type or destination cluster instantly updates the KPI cards and all four visualizations.

Purpose and Value

The purpose of this dashboard is to bring supplier activity and transportation performance into a single analytical view. Instead of examining individual records, the dashboard converts raw AIS data into KPIs and visualizations covering:

- Transportation activity
- Shipment volume
- Destination distribution
- Vessel-type transportation contribution
- Operational ETA performance
- Supplier / vessel-level performance

This supports supply-chain teams in identifying high-activity suppliers, understanding transportation patterns, and investigating destinations or transportation categories that warrant further analysis.

Limitations and Assumptions

The AIS dataset used for this project does not contain several conventional business fields typically required for a supplier and transportation dashboard, including actual supplier name, carrier name, shipment cost, fulfilment status, and actual delivery timestamp. Consequently, the dashboard uses clearly identified proxies based on available fields.

To address this, clearly defined operational proxies were used based on the fields that are available:

Business Concept	**Proxy Used**
Supplier	Vessel Name
Carrier	Vessel Type
Region / Route	Destination Cluster
Transportation Cost	Distance

Table 7: Business concepts and the proxies used

Tools Used

- **Microsoft Power BI** – Used to build the interactive dashboard, KPI cards, charts, slicers, and data visualizations [7].
- **Power Query** – Used for data import, data type checking, transformation, and preparation [9].
- **DAX (Data Analysis Expressions)** – Used to create calculated measures and KPI calculations such as Transportation Cost Proxy, Fulfillment Rate, On-Time Rate, and Supplier Quality Score [6, 10].

- **Kaggle** – Used to obtain the AIS Data Complete Weather dataset [4].
- **Google Colab** – Used to clean the raw AIS dataset before it was imported into Power BI.
- **Data Visualization** – Power BI charts such as bar/column charts and donut charts were used for supplier, carrier, route, and transportation analysis.

Summary

Milestone 3 extends the project from inventory and delivery analysis (Milestones 1 and 2) into supplier and transportation analytics. The dashboard delivers KPI monitoring, supplier-level analysis, transportation distribution, carrier / vessel-type analysis, and interactive filtering – providing a consolidated view that helps identify transportation patterns and operational performance areas.

Dashboard Output

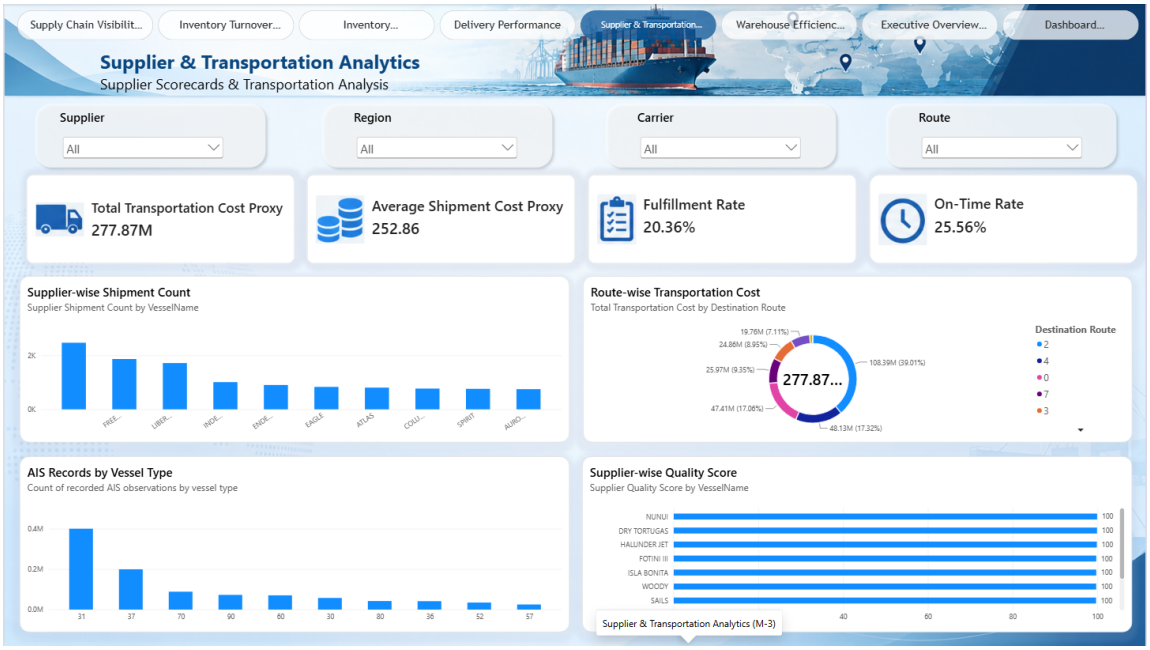


Figure 7: Page 5 – Supplier & Transportation Analytics dashboard

Milestone 4 – Warehouse Analytics & Final Dashboard

The main objectives of Milestone 4 were:

- Develop warehouse efficiency reports.
- Create an executive overview dashboard.
- Optimize dashboard performance.
- Carry out final testing, documentation, and deployment.

This milestone consolidates the data model, KPIs and dashboards built in Milestones 1–3 into a warehouse-efficiency view and a final executive-level summary, and closes out the project with performance optimization, testing and deployment.

Dashboard – Page 6: Warehouse Efficiency Reports

This page, titled *"Warehouse & Operational Efficiency Report"*, presents operational performance across the eight destination regions (`dest_cluster` 0–7).

KPI Cards

KPI	Result
Total AIS Records	1M
Cargo Movement Records	224K
Operational Utilization Proxy	20.36%
Average ETA (hours)	59.89

Table 8: Page 6 KPI cards

Cargo Movement Records by Destination

A horizontal bar chart shows cargo movement (moving AIS) records by `dest_cluster`. Cluster 2 leads with 97K records, followed by cluster 0 (34K) and cluster 4 (27K), while cluster 5 has the fewest (3K).

Operational Records by Vessel Type

A donut chart shows the distribution of operational records across vessel types, totalling 1M records. The largest vessel-type segment accounts for 36.43% (400K records), followed by 18.14% (199K) and 8.05% (88K), with several smaller vessel-type segments making up the remainder.

Wind Speed vs Vessel Speed

A scatter chart compares wind speed with average vessel speed (Average of `SOG_kmh`). The purpose of this visualization is exploratory: it allows the user to observe whether vessel speeds vary under different wind conditions. The chart should not be interpreted as proving a direct causal relationship between wind and vessel speed.

Average ETA by Destination

A bar chart shows average ETA (in the report's scaled units) by destination, with values ranging from 0.14 to 0.81. Cluster 0 shows the highest value (0.81), followed by cluster 6 (0.75) and cluster 4 (0.59), highlighting destinations with comparatively longer estimated arrival times.

Dashboard – Page 7: Executive Overview

This page, titled *"Executive Overview – Supply Chain Operational Performance at a Glance"*, consolidates the project's key metrics into a single executive-facing summary view.

KPI Cards

KPI	Result
Total Transportation Distance	277.87M
Average Speed (km/h)	4.36

Table 9: Page 7 KPI cards

Total Transportation Distance by Destination

A bar chart shows total transportation distance by destination, led by cluster 2 at 108M, followed by cluster 4 (48M) and cluster 0 (47M), with cluster 5 the lowest at 0M.

AIS Records by Destination Cluster

A column chart shows the AIS record distribution across the eight destination clusters, with cluster 2 dominating at 0.43M records and clusters 5 and 6 contributing the least (0.01M and 0.03M respectively).

AIS Records by Speed Category

A horizontal bar chart shows AIS records by speed category: Stopped records dominate at 0.88M, followed by Moderate (0.10M), Slow (0.08M) and Fast (0.04M).

Average Vessel Speed by Destination Cluster

A bar chart shows average vessel speed by destination cluster, ranging from 3.8 km/h (cluster 2) to 8.4 km/h (cluster 6).

Records by Vessel Type

A donut chart mirrors the vessel-type distribution seen on the Warehouse Efficiency page (1M total records), reused here to give executives a consistent view of vessel-type composition.

Filters

The Executive Overview page provides four report-level filters for interactive exploration:

- Date
- Destination
- Vessel Type
- Operational Status

Dashboard – Page 8: Operational KPI Summary

This page, titled *"Operational KPI Summary – Destination-wise Operational Performance"*, presents a consolidated matrix table combining the core operational KPIs for each destination, with conditional formatting (red–yellow–green) applied to highlight relative performance.

Destination-wise KPI Table

Destination	Operational Utilization (%)	On-Time Rate (%)	Avg. Vessel Speed (km/h)	Avg. ETA (hours)
0	22.60%	22.34%	5.16	60.27
1	15.71%	42.37%	3.88	55.46
2	22.33%	23.70%	3.79	60.25
3	17.66%	22.26%	4.80	61.31
4	15.01%	15.26%	4.63	64.00
5	29.37%	85.99%	6.22	27.84
6	32.10%	53.20%	8.40	44.81
7	25.16%	20.78%	4.28	63.76
Total	**20.36%**	**25.56%**	**4.36**	**59.89**

Table 10: Destination-wise operational KPI summary

The conditional formatting is applied to the Operational Utilization % and On-Time Rate columns. Destinations 5 and 6 show the highest utilization (29.37% and 32.10%), and destination 5 is the strongest on on-time rate (85.99%), whereas destinations 1 and 4 show the lowest utilization (15.71% and 15.01%) and destination 4 the lowest on-time rate (15.26%). This makes the table a useful at-a-glance diagnostic view for stakeholders.

Dashboard Performance Optimization

As part of finalizing the report, the following optimization practices were applied across all dashboard pages:

- Reused pre-aggregated DAX measures (from Milestones 1–3) across pages instead of creating duplicate calculations, reducing model complexity.
- Relied on the existing star-schema data model (`Fact_Vessel` with `dim_Vessel` and `dim_Destination`) to keep cross-page filtering efficient.

- Limited each dashboard page to a focused set of visuals and KPI cards to keep query and rendering load manageable.
- Used consistent slicers and filters (Date, Destination, Vessel Type, Movement/Operational Status, Speed Category) across pages so filter context is predictable and reusable.
- Reused common visuals (e.g. the Records by Vessel Type donut chart) across the Warehouse Efficiency and Executive Overview pages rather than recomputing them separately.

Final Testing, Documentation & Deployment

Testing

Each dashboard page and its slicers were tested to confirm that KPI cards, charts and the Operational KPI Summary table respond correctly to filter selections (Date, Destination, Vessel Type, Operational Status, Speed Category), and that totals on the summary table reconcile with the KPI cards shown on other pages.

Documentation

Milestone-wise documentation (Milestones 1–4) was prepared covering the data model, DAX measures, dashboard pages, KPIs, visualizations and business insights, providing a complete technical record of the project.

Deployment

The finalized Power BI report, consisting of eight dashboard pages linked by a common navigation bar – Supply Chain Visibility, Inventory Turnover Analysis, Inventory Performance, Delivery Performance, Supplier & Transportation Analytics, Warehouse Efficiency Reports, Executive Overview and Operational KPI Summary – was published as the final deliverable for the Supply Chain Visibility System with Optimization Analytics project.

Key Dashboard Results

- Total AIS records analyzed: approximately 1 million, across ∼13,000 unique vessels.
- Cargo movement (moving) records: approximately 224K.
- Overall operational utilization proxy: 20.36%, ranging from 15.01% to 32.10% across destinations.
- Overall on-time rate: 25.56%, ranging from 15.26% to 85.99% across destinations.
- Overall average vessel speed: 4.36 km/h, ranging from 3.79 to 8.40 km/h across destinations.
- Overall average ETA: 59.89 hours, ranging from 27.84 to 64.00 hours across destinations.
- Total transportation distance: approximately 277.87 million km, led by destination cluster 2 (108M km).
- Destination cluster 5 recorded the strongest on-time performance (85.99%); destination cluster 4 the weakest (15.26%).

Business Insights

1. **Warehouse & Operational Utilization** – Utilization varies widely by destination (15%–32%), indicating uneven operational load and potential for rebalancing across regions.
2. **On-Time Performance Gaps** – The gap between the best-performing destination (85.99% on-time) and the weakest (15.26%) points to specific regions that would benefit most from targeted delivery-process improvements.
3. **Speed vs. ETA Trade-off** – Destinations with higher average vessel speed (e.g. cluster 6 at 8.40 km/h) tend to show shorter average ETA, reinforcing the operational link between vessel speed and estimated arrival time.
4. **Executive Visibility** – The Executive Overview page consolidates total distance, average speed, destination-level activity, speed-category distribution and vessel-type composition into a single glanceable view suitable for leadership reporting.

Dashboard Output

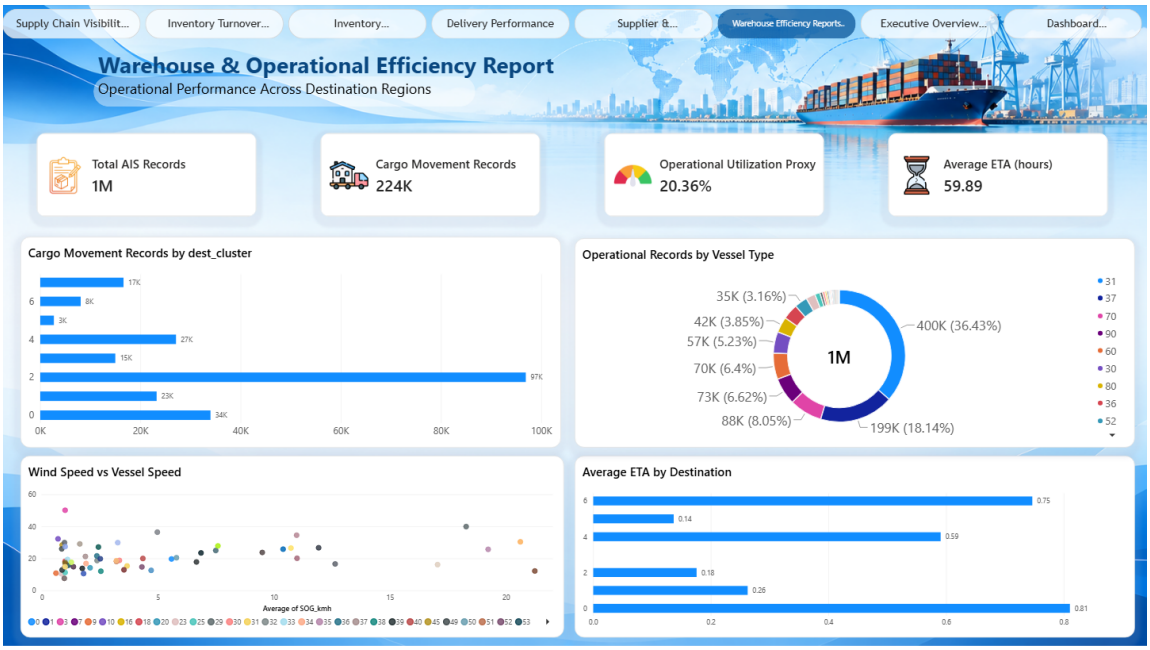
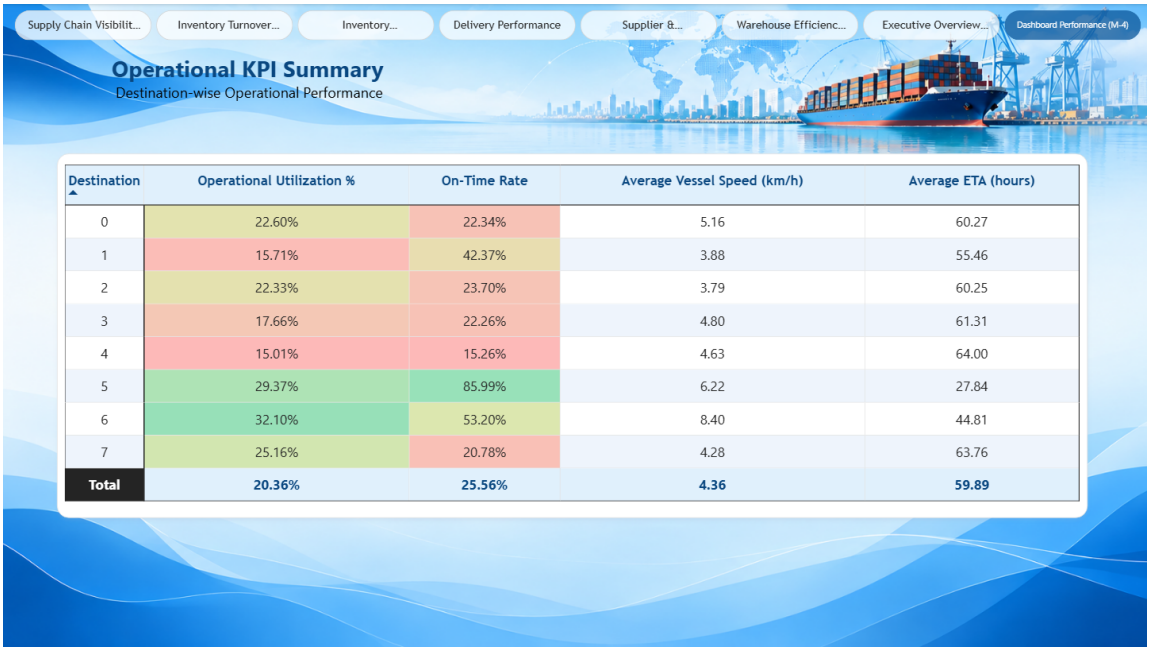


Figure 8: Page 6 – Warehouse Efficiency Reports



Destination	Operational Utilization %	On-Time Rate	Average Vessel Speed (km/h)	Average ETA (hours)
0	22.60%	22.34%	5.16	60.27
1	15.71%	42.37%	3.88	55.46
2	22.33%	23.70%	3.79	60.25
3	17.66%	22.26%	4.80	61.31
4	15.01%	15.26%	4.63	64.00
5	29.37%	85.99%	6.22	27.84
6	32.10%	53.20%	8.40	44.81
7	25.16%	20.78%	4.28	63.76
Total	**20.36%**	**25.56%**	**4.36**	**59.89**

Figure 9: Page 7 – Executive Overview

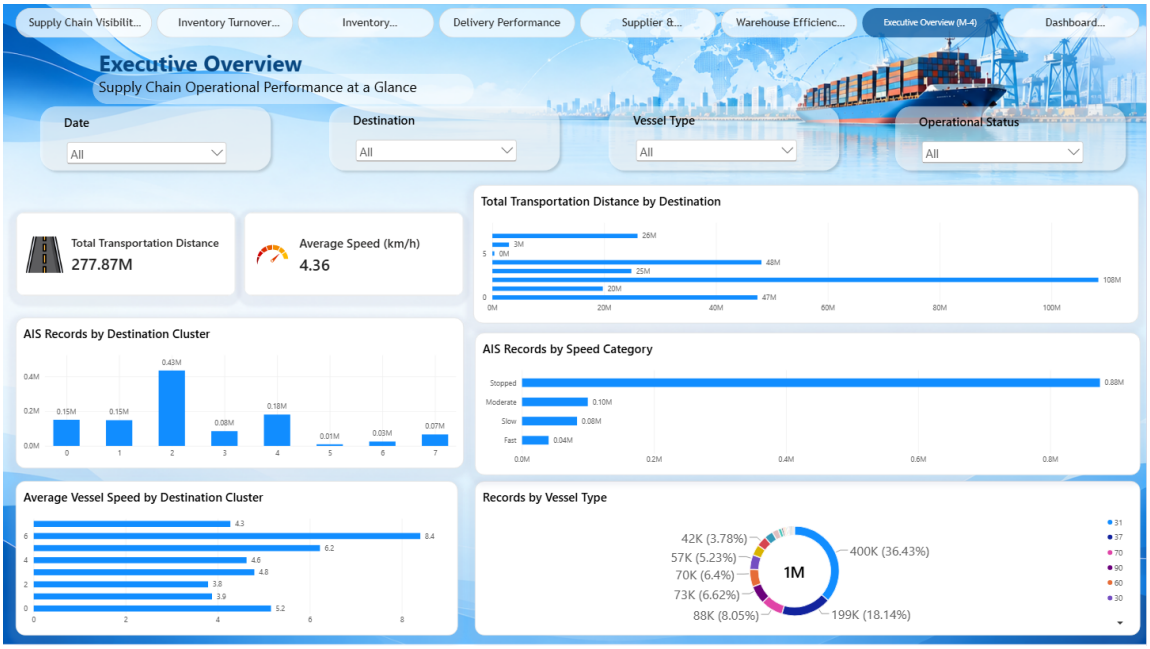


Figure 10: Page 8 – Operational KPI Summary

Conclusion

Milestone 4 completed the Supply Chain Visibility System with Optimization Analytics project by adding warehouse efficiency reporting and an executive-level overview on top of the data model and KPIs established in the earlier milestones. The Warehouse Efficiency Reports page consolidated cargo movement, operational utilization, vessel-type composition, wind-versus-vessel-speed context and ETA by destination, while the Executive Overview page brought transportation distance, speed and record distribution together into a single high-level summary for leadership. The Operational KPI Summary table provided a consolidated, conditionally formatted reference view of utilization, on-time rate, speed and ETA across all destinations.

With dashboard performance optimized, all pages tested against their filters and slicers, and full milestone-wise documentation completed, the final Power BI report was published as the project's deliverable, providing an end-to-end, interactive view of supply-chain visibility, inventory turnover, delivery performance, supplier and transportation analytics, and warehouse efficiency.

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